

**A**

$$\min_{u_{t+1}, \dots, u_{t+H}} \sum_{h=1}^H (\hat{y}_{t+h} - r)^2 + \rho \sum_{h=1}^H (u_{t+h} - u_{ss})^2 + \lambda \sum_{h=1}^H (\Delta u_{t+h})^2$$

s.t.  $\hat{y}_{t+h} = \text{VARX}(z_t, \dots, z_{t-L+1}; u_{t+h}, \dots)$ ,  $0 \leq u \leq u_{\max}$

$H = 1 \text{ s}$ ,  $r = -5\%$ ,  $\rho = 10^{-3}$ ,  $\lambda = 0.1$

